

Weekly Progress Report – Week 2

Course: Software Development

Project: Handwritten Recognition Game

Language: Java (Swing GUI)

Week: 2

Objective of Week 2

The main objective of this week was to build the foundation of the graphical user interface (GUI) for the Handwritten Recognition Game. During this phase, I focused on learning Java Swing components and creating the initial Home Page of the application.

Tasks Completed

1. Java Swing Fundamentals

I studied the basic concepts of Java Swing, which is used to create graphical user interfaces in Java. The following components were explored:

- JFrame
- JPanel
- JLabel
- JButton
- Layout Managers

This knowledge helped in designing the application's user interface.

2. Event Handling Concepts

I learned how Java handles user interactions through event-driven programming.

The following concepts were practiced:

- ActionListener
- Button Click Events
- Event Registration

- Event Processing

These concepts will be used later for page navigation and game controls.

3. Main Application Window Development

A main application window was created using JFrame.

Features implemented:

- Window title
- Window size configuration
- Center screen positioning
- Application close operation

This serves as the base window for the entire game.

4. GUI Components Integration

Different Swing components were added to the interface:

- Title Label
- Start Game Button
- Instructions Button
- Exit Button
- Organized Panels

These components create an interactive and user-friendly environment.

5. Home Page Design

The Home Page was designed as the first screen of the game.

Home Page Features:

- Project Title Display
- Welcome Message
- Navigation Buttons
- Clean and Simple Layout
- User-Friendly Interface

The design focuses on simplicity and ease of use.

Challenges Faced

- Understanding Swing layout management.
- Managing component positioning.
- Learning event-driven programming concepts.

These challenges were resolved through practice and testing.

Skills Learned

- Java Swing GUI Development
 - Event Handling
 - JFrame and JPanel Management
 - Button Actions
 - Basic UI Design Principles
-

Result

Successfully developed the basic GUI structure and Home Page of the Handwritten Recognition Game. The application is now ready for Week 3 development, where additional pages (Instruction Page, Game Page, and Result Page) will be created and connected.

Files Prepared for Upload

✓ Home Page Source Code

Code:

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
import java.awt.event.ActionEvent;
```

```
import java.awt.event.ActionListener;
```

```
public class HomePage extends JFrame {

    public HomePage() {

        setTitle("Handwritten Recognition Game");

        setSize(900, 600);

        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        setLocationRelativeTo(null);

        // Main panel

        JPanel panel = new JPanel();

        panel.setLayout(new BorderLayout());

        // Title

        JLabel title = new JLabel(

            "WELCOME TO HANDWRITTEN RECOGNITION GAME",

            SwingConstants.CENTER);

        title.setFont(new Font("Arial", Font.BOLD, 26));

        // Button Panel

        JPanel buttonPanel = new JPanel();

        buttonPanel.setLayout(new GridLayout(3, 1, 15, 15));
```

```
    JButton startButton = new JButton("START GAME");

    JButton instructionButton = new JButton("INSTRUCTIONS");

    JButton exitButton = new JButton("EXIT");
    buttonPanel.add(startButton);

    buttonPanel.add(instructionButton);

    buttonPanel.add(exitButton);


    panel.add(title, BorderLayout.NORTH);

    panel.add(buttonPanel, BorderLayout.CENTER);


    add(panel);


    // EVENTS

    startButton.addActionListener(e -> {

        JOptionPane.showMessageDialog(this, "Game Page Open");

    });

    instructionButton.addActionListener(e -> {

        JOptionPane.showMessageDialog(this, "Instruction Page Open");

    });

    exitButton.addActionListener(e -> {

        System.exit(0);
```

```
});
```

```
setVisible(true);
```

```
}
```

```
public static void main(String[] args) {
```

```
    new HomePage();
```

```
}
```

```
}
```

✓ Home Page Screenshot:



✓ Weekly Progress Report:

WEEK 2 PROGRESS REPORT

Project Name: Handwritten Recognition Game

Course: Software Development

Programming Language: Java

Week Number: 2

Objectives:

1. Learn Java Swing Basics
2. Learn Event Handling Concepts
3. Create Main Application Window
4. Add Buttons, Labels, and Panels
5. Design the Home Page

Work Completed:

During Week 2, Java Swing fundamentals were studied and implemented. Various GUI components such as JFrame, JPanel, JLabel, and JButton were explored and used to build the user interface. Event handling concepts were learned to manage button click actions.

A Home Page for the Handwritten Recognition Game was successfully designed and developed. The Home Page contains a welcome title along with three functional buttons: Start Game, Instructions, and Exit. The graphical user interface was tested to ensure proper layout and functionality.

Tools and Technologies Used:

- Java JDK
- Java Swing
- IntelliJ IDEA / Eclipse
- GitHub

Testing Performed:

- Main Window Loading Test – PASS
- Button Display Test – PASS
- Layout Alignment Test – PASS
- Event Handling Test – PASS

Problems Faced:

- GUI component alignment issues
- Layout management difficulties

Solutions Applied:

- Used appropriate Layout Managers
- Adjusted component spacing and positioning

Outcome:

The Home Page was successfully completed and is functioning correctly. All planned tasks for Week 2 were achieved according to the project work plan.

Status:

WEEK 2 SUCCESSFULLY COMPLETED